OLA Operational & Predictive Data Analysis Report:

1. Executive Summary

This project delivers a comprehensive, end-to-end data analysis of one month of operational ride data for **OLA** (Bangalore based 20k data). By bridging **descriptive business intelligence (SQL & Power BI)** with **predictive machine learning (Python & Scikit-Learn)**, this analysis uncovers critical performance metrics, operational bottlenecks, revenue drivers, and actionable optimization strategies.

Key high-level metrics evaluated from the dataset of **20,407 total bookings** include an overall ride success rate of **62.00%**, total revenue generation exceeding **₹11.14 Million**, and an overall unfulfilled/cancellation rate of **38.00%**.

2. Key Performance Indicators (KPIs) & Findings

A. Booking Status Distribution

- **Successful Rides**: 12,652 bookings (**62.00%**)
- **Cancelled by Driver**: 3,654 bookings (**17.91%**)
- **Cancelled by Customer**: 2,081 bookings (**10.20%**)
- **Driver Not Found**: 2,020 bookings (**9.90%**)
- *Observation*: Cancellations and unfulfilled bookings represent over **38%** of total demand, highlighting an urgent need for dispatch efficiency and driver-retention policies.

B. Revenue & Vehicle Performance

- **Total Revenue**: ₹11,148,671 across all vehicle segments.
- **Top Revenue-Generating Vehicle**: **Prime Sedan** generated the highest overall booking value (~₹1.67M), followed closely by **eBike** (~₹1.62M) and **Prime SUV** (~₹1.58M).
- **Payment Preferences**: **Cash** remains the dominant payment method (contributing ~₹3.82M), followed closely by **UPI** (~₹2.76M), while Credit and Debit cards account for a minor share.

C. Customer & Driver Satisfaction

- **Customer Ratings**: Averaged a consistent **3.99 / 5.0** across vehicle categories.
- **Driver Ratings**: Averaged **4.00 / 5.0**, indicating generally stable service quality despite high cancellation volumes.

3. Machine Learning Insights (Decision Tree Classifier)

To move beyond historical reporting, a **Decision Tree Classifier** was built using Python (scikit-learn) to predict and analyse the root causes of ride cancellations.

- **Model Objective**: Predict whether a ride results in a Success vs. Cancelled status based on operational features (Vehicle_Type, Payment_Method, V_TAT, C_TAT, Ride_Distance).
- **Key Decision Rules Discovered**:
    - **Rule 1 (Wait Time Thresholds)**: When Vehicle Turnaround Time (V_TAT) exceeds critical operational limits (e.g., >164 minutes in experimental splits), the probability of trip failure increases exponentially.
    - **Rule 2 (Customer Patience Limits)**: Customer cancellations spike when wait times and coordination delays surpass threshold tolerances, regardless of vehicle category.

4. Strategic Solutions & Recommendations

1. **Optimize Dispatch & Reduce Wait Times (V_TAT)**:
    - Implement real-time driver proximity matching to minimize pickup turnaround times, directly targeting the primary driver cancellation trigger.
2. **Dynamic Incentivization for Drivers**:
    - Introduce surge incentives or completion bonuses in high-cancellation zones to reduce personal/car-related driver cancellations.
3. **Promote Digital Payments (UPI)**:
    - Incentivize UPI transactions to reduce cash-handling friction, streamline trip closure, and speed up turnaround times.
4. **Predictive Fleet Allocation**:
    - Utilize time-series forecasting and decision tree risk triggers to pre-allocate fleets in high-demand micro-markets before bottlenecks occur.